

## CSS Classes

### Question 1

#### index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Pseudo-Classes</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <h1>Pseudo-Classes Demo</h1>
  </header>
  <main class="examples">
    <section>
      <h2>:hover</h2>
      <a href="#">Hover over this link</a>
      <p>The link changes color when the mouse is over it.</p>
    </section>
    <section>
      <h2>:focus</h2>
      <input type="text" placeholder="Click to focus">
      <p>The input gets an outline when it is focused.</p>
    </section>
    <section>
      <h2>:active</h2>
      <button>Press and hold</button>
      <p>The button changes background while being clicked.</p>
    </section>
    <section>
      <h2>:nth-child()</h2>
      <ul>
        <li>Item 1</li>
        <li>Item 2</li>
        <li>Item 3</li>
        <li>Item 4</li>
      </ul>
      <p>Every even list item has a gray background.</p>
    </section>
  </main>
</body>
</html>
```

#### style.css

```
body {
  margin: 0;
  font-family: Arial, sans-serif;
  text-align: center;
}
```

```

header {
  background: #333;
  color: white;
  padding: 15px;
}
.examples {
  display: flex;
  justify-content: center;
  gap: 30px;
  padding: 40px;
}
section {
  width: 200px;
  padding: 15px;
  border: 1px solid #ccc;
  border-radius: 8px;
}
a {
  color: blue;
}
a:hover {
  color: red;
}
input:focus {
  outline: 3px solid orange;
}
button {
  padding: 10px 20px;
  background: steelblue;
  color: white;
  border: none;
}
button:active {
  background: darkblue;
}
ul {
  list-style: none;
  padding: 0;
}
li {
  padding: 5px;
}
li:nth-child(even) {
  background: lightgray;
}

```

## Question 2

### index.html

```

<!DOCTYPE html>
<html lang="en">

```

```

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Tooltips</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main class="container">
    <span class="tip">Home
      <span class="tooltip">Go to home page</span>
    </span>
    <span class="tip">Settings
      <span class="tooltip">Change your settings</span>
    </span>
    <span class="tip">Help
      <span class="tooltip">Get help and support</span>
    </span>
  </main>
</body>
</html>

```

### **style.css**

```

body {
  margin: 0;
  font-family: Arial, sans-serif;
}
.container {
  height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  gap: 60px;
}
.tip {
  position: relative;
  padding: 10px 20px;
  background: steelblue;
  color: white;
  border-radius: 6px;
  cursor: pointer;
}
.tooltip {
  position: absolute;
  bottom: 120%;
  left: 50%;
  transform: translateX(-50%);
  width: 150px;
  padding: 8px;
  background: #333;
  color: white;
  text-align: center;
  border-radius: 6px;
}

```

```
font-size: 14px;
opacity: 0;
transition: 0.3s ease-in-out;
}
.tip:hover .tooltip {
  opacity: 1;
}
```

### Question 3

#### index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Image Showcase</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1>Image Showcase</h1>
  <main class="gallery">
    <figure>
      <div class="img-box"></div>
      <figcaption>Mountains</figcaption>
    </figure>
    <figure>
      <div class="img-box"></div>
      <figcaption>Beach</figcaption>
    </figure>
    <figure>
      <div class="img-box"></div>
      <figcaption>Forest</figcaption>
    </figure>
    <figure>
      <div class="img-box"></div>
      <figcaption>Desert</figcaption>
    </figure>
    <figure>
      <div class="img-box"></div>
      <figcaption>City</figcaption>
    </figure>
    <figure>
      <div class="img-box"></div>
      <figcaption>River</figcaption>
    </figure>
  </main>
</body>
</html>
```

#### style.css

```
body {
```

```

margin: 0;
padding: 20px;
font-family: Arial, sans-serif;
text-align: center;
}
.gallery {
display: grid;
grid-template-columns: repeat(3, 1fr);
gap: 20px;
}
figure {
margin: 0;
}
.img-box {
position: relative;
border-radius: 10px;
box-shadow: 0 4px 10px gray;
overflow: hidden;
}
.img-box img {
width: 100%;
height: 200px;
display: block;
transition: transform 0.3s ease;
}
.img-box::after {
content: "";
position: absolute;
top: 0;
left: 0;
width: 100%;
height: 100%;
background: rgba(0, 0, 0, 0);
transition: background 0.3s ease;
}
.img-box:hover img {
transform: scale(1.1);
}
.img-box:hover::after {
background: rgba(0, 0, 0, 0.4);
}
figcaption {
padding: 10px;
font-weight: bold;
}
}
@media (max-width: 700px) {
.gallery {
grid-template-columns: 1fr;
}
}
}

```

## Question 4

### index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>CSS Selectors</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <h1>CSS Selectors Practice</h1>
  </header>
  <main>
    <p>This paragraph is styled by the element selector.</p>
    <p class="highlight">This paragraph uses a class selector.</p>
    <p id="special">This paragraph uses an ID selector.</p>
    <a href="https://example.com">This link uses an attribute selector</a>
  <section>
    <h2>Combinators</h2>
    <p>Direct child of section (child selector).</p>
    <ul>
      <li>First item</li>
      <li>Second item (adjacent sibling)</li>
      <li>Third item (general sibling)</li>
      <li>Fourth item (general sibling)</li>
    </ul>
    <div>
      <p>Paragraph inside div (descendant selector).</p>
    </div>
  </section>
</main>
<footer>
  <p>Selectors Practice Page</p>
</footer>
</body>
</html>
```

### style.css

```
body {
  margin: 0;
  font-family: Arial, sans-serif;
}
header, footer {
  background: #333;
  color: white;
  text-align: center;
  padding: 15px;
}
main {
  padding: 30px;
```

```

}
p {
  margin: 10px 0;
}
.highlight {
  background: yellow;
}
#special {
  color: darkred;
  font-weight: bold;
}
a[href^="https"] {
  color: green;
}
section > p {
  border-left: 4px solid steelblue;
  padding-left: 10px;
}
li + li {
  color: blue;
}
li ~ li {
  font-style: italic;
}
div p {
  background: lightgray;
  padding: 10px;
}

```

## Question 5

### index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Button Collection</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main class="buttons">
    <button class="primary">Primary</button>
    <button class="secondary">Secondary</button>
    <button class="outline">Outline</button>
    <button class="primary" disabled>Disabled</button>
  </main>
</body>
</html>

```

### style.css

```

body {

```

```
margin: 0;
font-family: Arial, sans-serif;
}
.buttons {
  height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  gap: 20px;
}
button {
  padding: 12px 25px;
  font-size: 16px;
  border: none;
  border-radius: 8px;
  box-shadow: 0 3px 6px gray;
  cursor: pointer;
  transition: all 0.3s ease;
}
.primary {
  background: steelblue;
  color: white;
}
.secondary {
  background: gray;
  color: white;
}
.outline {
  background: white;
  color: steelblue;
  border: 2px solid steelblue;
}
button:hover {
  transform: translateY(-3px);
  box-shadow: 0 6px 12px gray;
}
button:active {
  transform: translateY(1px);
  box-shadow: 0 1px 3px gray;
}
button:focus {
  outline: 3px solid orange;
}
button:disabled {
  background: lightgray;
  color: darkgray;
  box-shadow: none;
  cursor: not-allowed;
  transform: none;
}
```

## Question 6

### index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>CSS Animations</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main class="stage">
    <div class="box">Moving Box</div>
    <div class="ball"></div>
    <h2 class="text">Color Changing Text</h2>
  </main>
</body>
</html>
```

### style.css

```
body {
  margin: 0;
  font-family: Arial, sans-serif;
}
.stage {
  height: 100vh;
  display: flex;
  flex-direction: column;
  justify-content: center;
  align-items: center;
  gap: 50px;
}
.box {
  width: 120px;
  padding: 15px;
  background: steelblue;
  color: white;
  text-align: center;
  animation: move 2s ease-in-out infinite alternate;
}
.ball {
  width: 50px;
  height: 50px;
  background: tomato;
  border-radius: 50%;
  animation: bounce 1s ease-in-out infinite alternate;
  animation-delay: 0.5s;
}
.text {
  animation: colorChange 3s ease-in-out infinite;
  animation-iteration-count: infinite;
  animation-direction: alternate;
```

```

}
@keyframes move {
  from {
    transform: translateX(-150px);
  }
  to {
    transform: translateX(150px);
  }
}
@keyframes bounce {
  from {
    transform: translateY(0);
  }
  to {
    transform: translateY(-60px);
  }
}
@keyframes colorChange {
  0% {
    color: red;
  }
  50% {
    color: green;
  }
  100% {
    color: blue;
  }
}

```

## Question 7

### index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>UI Components</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main>
    <section class="card">
      <h1>UI Components</h1>
      <input type="text" placeholder="Search...">
      <select>
        <option>Select Category</option>
        <option>Books</option>
        <option>Electronics</option>
        <option>Clothes</option>
      </select>
    </section>
  </main>
</body>
</html>

```

```

    <div class="actions">
      <button class="primary">Search</button>
      <button class="secondary">Reset</button>
    </div>
  </section>
</main>
</body>
</html>

```

### style.css

```

body {
  margin: 0;
  font-family: Arial, sans-serif;
  background: linear-gradient(to bottom right, #e0f7fa, #fce4ec);
}
main {
  height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
}
.card {
  width: 320px;
  padding: 30px;
  background: white;
  border-radius: 12px;
  box-shadow: 0 6px 15px gray;
  display: flex;
  flex-direction: column;
  gap: 15px;
  text-align: center;
}
input, select, button {
  padding: 10px;
  border: 1px solid #ccc;
  border-radius: 8px;
  font-size: 15px;
}
input:focus, select:focus {
  outline: none;
  border-color: steelblue;
  box-shadow: 0 0 5px steelblue;
}
.actions {
  display: flex;
  gap: 10px;
}
.actions button {
  flex: 1;
  cursor: pointer;
  transition: all 0.3s ease;
}

```

```

.primary {
  background: steelblue;
  color: white;
}
.secondary {
  background: lightgray;
}
button:hover {
  opacity: 0.8;
  transform: scale(1.05);
}

```

## Question 8

### index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Box-Sizing Comparison</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main>
    <h1>Box-Sizing Comparison</h1>
    <section class="boxes">
      <div class="box default">Default (content-box)</div>
      <div class="box border">border-box</div>
    </section>
    <p>Both boxes have width 250px and padding 30px. With the default content-box, padding and border are added outside the width, so the box becomes bigger. With border-box, padding and border are included inside the width, so the total size stays 250px.</p>
  </main>
</body>
</html>

```

### style.css

```

body {
  margin: 0;
  font-family: Arial, sans-serif;
}
main {
  min-height: 100vh;
  display: flex;
  flex-direction: column;
  justify-content: center;
  align-items: center;
  text-align: center;
  padding: 20px;
}

```

```

.bboxes {
  display: flex;
  gap: 30px;
  flex-wrap: wrap;
  justify-content: center;
}
.box {
  width: 250px;
  padding: 30px;
  border: 5px solid #333;
  color: white;
}
.default {
  box-sizing: content-box;
  background: tomato;
}
.border {
  box-sizing: border-box;
  background: steelblue;
}
p {
  max-width: 600px;
  margin-top: 30px;
  line-height: 1.6;
}

```

## Question 9

### index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Combined Effects</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main class="effects">
    <button class="grow">Grow</button>
    <button class="rotate">Rotate</button>
    <button class="lift">Lift</button>
    <button class="color">Color</button>
  </main>
</body>
</html>

```

### style.css

```

body {
  margin: 0;
  font-family: Arial, sans-serif;
}

```

```

.effects {
  height: 100vh;
  display: flex;
  justify-content: space-evenly;
  align-items: center;
}
button {
  padding: 15px 30px;
  font-size: 16px;
  border: none;
  border-radius: 8px;
  background: steelblue;
  color: white;
  box-shadow: 0 3px 6px gray;
  cursor: pointer;
  transition: all 0.3s ease;
}
.grow:hover {
  transform: scale(1.1);
  background: darkblue;
}
.rotate:hover {
  transform: rotate(5deg);
  background: tomato;
}
.lift:hover {
  transform: translateY(-8px);
  box-shadow: 0 10px 20px gray;
}
.color:hover {
  background: seagreen;
  transform: scale(1.05);
}
button:active {
  transform: scale(0.95);
}

```

## Question 10

### index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Mini UI Layout</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <h1>My App</h1>

```

```

</header>
<main>
  <section class="card">
    
    <h2>Create Account</h2>
    <form>
      <input type="text" placeholder="Full Name">
      <input type="email" placeholder="Email">
      <input type="password" placeholder="Password">
      <div class="actions">
        <button type="submit">Sign Up</button>
        <button type="reset" class="secondary">Clear</button>
      </div>
    </form>
  </section>
</main>
<footer>
  <p>&copy; 2026 My App</p>
</footer>
</body>
</html>

```

### style.css

```

* {
  box-sizing: border-box;
}
body {
  margin: 0;
  font-family: Arial, sans-serif;
  background: #f4f4f4;
  display: flex;
  flex-direction: column;
  min-height: 100vh;
}
header, footer {
  background: #333;
  color: white;
  text-align: center;
  padding: 15px;
}
main {
  flex: 1;
  display: flex;
  justify-content: center;
  align-items: center;
  padding: 20px;
}
.card {
  width: 100%;
  max-width: 350px;
  padding: 30px;
  background: white;

```

```
border: 1px solid #ddd;
border-radius: 12px;
box-shadow: 0 6px 15px gray;
text-align: center;
}
.card img {
border-radius: 50%;
border: 3px solid steelblue;
}
form {
display: flex;
flex-direction: column;
gap: 12px;
}
input {
padding: 10px;
border: 1px solid #ccc;
border-radius: 6px;
}
input:focus {
outline: none;
border-color: steelblue;
box-shadow: 0 0 5px steelblue;
}
.actions {
display: flex;
gap: 10px;
}
button {
flex: 1;
padding: 10px;
border: none;
border-radius: 6px;
background: steelblue;
color: white;
cursor: pointer;
transition: all 0.3s ease;
}
.secondary {
background: gray;
}
button:hover {
opacity: 0.85;
transform: translateY(-2px);
}
```